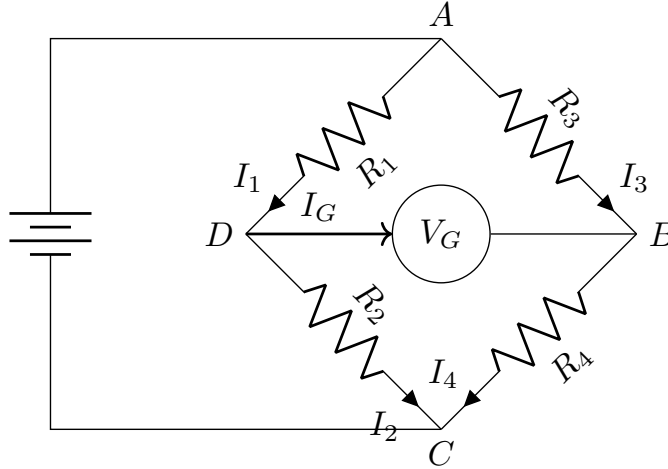


Problem Set - 1 (Answer Key)

Question 1

Part (a)



Let I_G denote the current through the Galvanometer. Using Kirchhoff's Current Law (KCL) at node D , we get :

$$I_1 = I_2 + I_G \quad (1)$$

Similarly, at node B , we get :

$$I_3 + I_G = I_4 \quad (2)$$

Consider the loops $A \rightarrow D \rightarrow G \rightarrow B \rightarrow A$ and $C \rightarrow D \rightarrow G \rightarrow B \rightarrow C$ as 1 and 2, respectively. Using Kirchhoff's Voltage Law on Loop, we get

$$I_1 R_1 + I_G R_G - I_3 R_3 = 0 \quad (3)$$

And for Loop 2

$$-I_2 R_2 + I_G R_G + I_4 R_4 = 0 \quad (4)$$

Now, during Balance/Null condition, $I_G = 0$

Therefore, Equations (3) and (4) simplify to :

$$I_1 R_1 = I_3 R_3$$

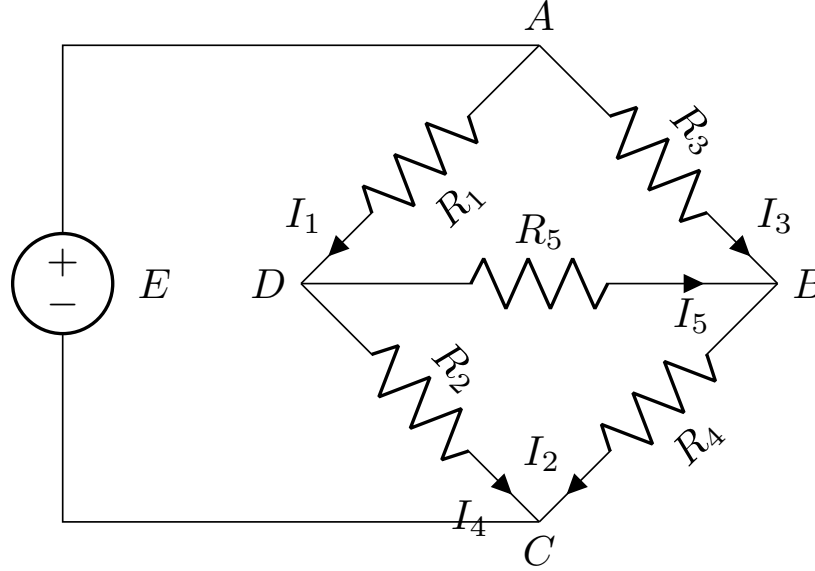
and

$$I_4 R_4 = I_2 R_2$$

From (1) and (2), we get $I_1 = I_2$ and $I_3 = I_4$. Using these conditions on the above equations, we get the following balancing condition :

$$\frac{R_1}{R_2} = \frac{R_3}{R_4} \quad (5)$$

Part (b)



Consider Equations (1) to (4) (replacing R_G with R_5). We approach this problem by finding 3 equations, in terms of 3 current variables. For instance, we take I_1 , I_3 and I_G as the three variables; x, y and z respectively. Then, Eqn (3) becomes :

$$R_1x + R_5z - R_3y = 0 \quad (6)$$

Since Eqn (4) has terms dependant on I_2 and I_4 , we use Eqns (1) and (2) to remove the dependancy as :

$$R_2x - R_4y - (R_2 + R_4 + R_5)z = 0 \quad (7)$$

For the last equation, consider the loop $A \rightarrow E \rightarrow C \rightarrow D \rightarrow A$. Using KVL, and Eqn (2), we get :

$$0x + (R_3 + R_4)y + R_4z - V = 0 \quad (8)$$

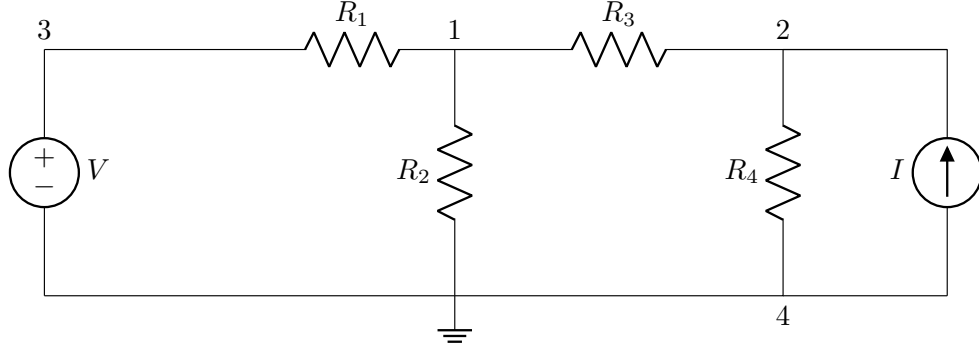
Equations (6), (7) and (8) are a system of three linear equations, that can be solved using various methods. Upon solving, we get the final expression :

$$I_5 = \frac{(R_2R_3 - R_4R_1)V}{R_1R_3(R_2 + R_4 + R_5) + R_2R_4(R_1 + R_3 + R_5) + R_5(R_1R_4 + R_2R_3)} \quad (9)$$

Note that if $I_5 = 0$, we get the balancing condition as in Eqn. (5).

Question 2

Part (a)



For convenience, we use conductance G . Using Node Voltage Theorem (and KCL) on node 1, we get :

$$\begin{aligned} G_1(V - V_1) + G_3(V_2 - V_1) - G_2V_1 &= 0 \\ \implies -V_1(G_1 + G_2 + G_3) + G_3V_2 + G_1V &= 0 \end{aligned} \quad (10)$$

Similarly in node 2, we get :

$$-G_3V_1 + (G_3 + G_4)V_2 = I \quad (11)$$

We now have two linear equations w.r.t variables V_1 and V_2 . We solve for them using preferred methods. The final solution obtained would be :

$$V_1 = \frac{G_1V(G_3 + G_4) + G_3I}{(G_1 + G_2)(G_3 + G_4) + G_3G_4} \quad (12)$$

$$V_2 = \frac{G_1(G_3V + I) + I(G_2 + G_3)}{(G_1 + G_2)(G_3 + G_4) + G_3G_4} \quad (13)$$

Voltage across R_3 can be acquired by evaluating $V_1 - V_2$, which gives us,

$$V_{R_3} = \frac{G_1G_4V - (G_1 + G_2)I}{(G_1 + G_2)(G_3 + G_4) + G_3G_4} \quad (14)$$

$$= \frac{R_3(VR_2 - (R_1 + R_2)R_4I)}{(R_1 + R_2)(R_3 + R_4) + R_1R_2} \quad (15)$$

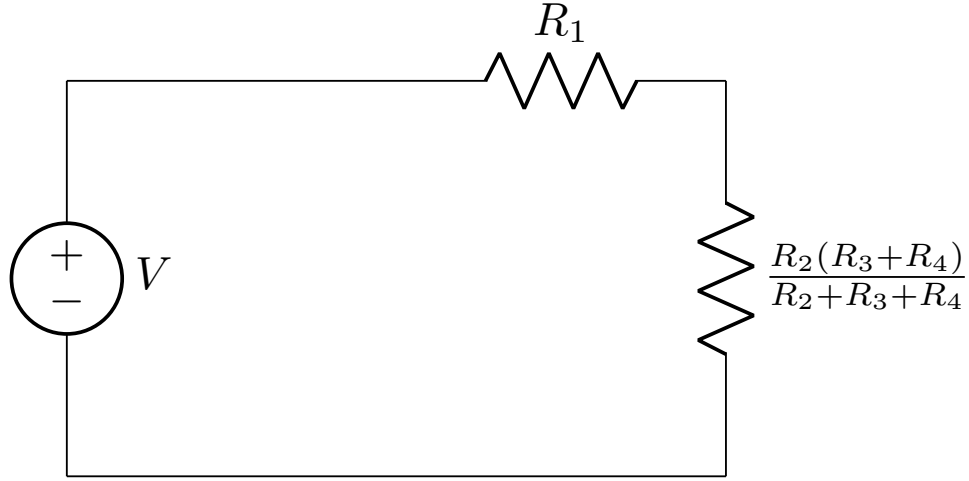
$$I_{R_3} = \frac{VR_2 - (R_1 + R_2)R_4I}{(R_1 + R_2)(R_3 + R_4) + R_1R_2} \quad (16)$$

If $VR_2 > (R_1 + R_2)R_4I$, I_{R_3} is positive, hence current flows from node 1 to 2, and when $VR_2 < (R_1 + R_2)R_4I$, I_{R_3} is negative, hence current flows from node 2 to 1.

Part (b)

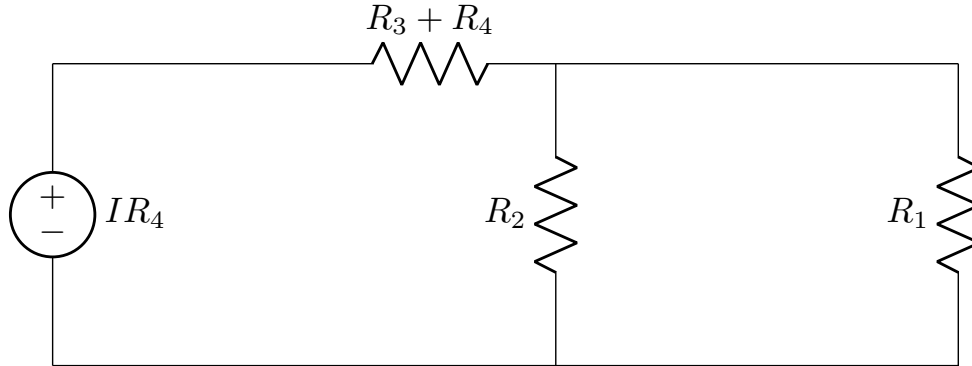
First, we calculate the contribution of the voltage source on R_1 , by opening the current

source. Using Thevenin's theorem, the equivalent circuit can be constructed as given below.



$$\Rightarrow V_v = \frac{R_1}{R_1 + \frac{R_2(R_3+R_4)}{R_2+R_3+R_4}} V \quad (17)$$

After shorting the voltage, we measure the current source contribution on R_1 . A similar equivalent circuit can be constructed as such,



$$\Rightarrow V_i = \frac{IR_4 R_1 \parallel R_2}{R_3 + R_4 + R_1 \parallel R_2}, \quad R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2} \quad (18)$$

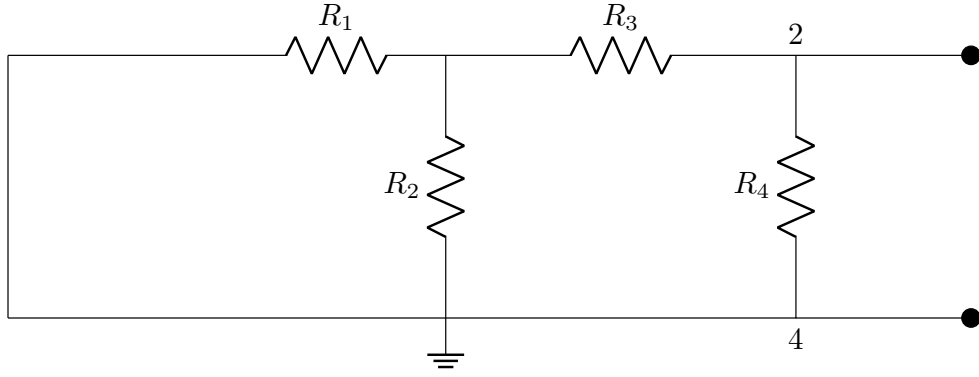
Therefore,

$$V_{R_1} = V_v + V_i = \frac{V R_1}{R_1 + \frac{R_2(R_3+R_4)}{R_2+R_3+R_4}} + \frac{I R_4 R_1 \parallel R_2}{R_3 + R_4 + R_1 \parallel R_2} \quad (19)$$

$$V_{R_1} = \frac{V R_1 (R_2 + R_3 + R_4) + I R_4 R_1 R_2}{(R_1 + R_2)(R_3 + R_4) + R_1 R_2} \quad (20)$$

Part (c)

According to Thevenin's Theorem (or corollary), the voltage of the equivalent circuit is same as the measured voltage across the nodes. Hence, Thevenin Voltage as seen



from Terminal 2 to 4, $V_{Th} = V_2$ (given in Eqn. (13)). Upon removing the sources by shorting/opening, the equivalent circuit we get is,

$$R_{Th} = (R_3 + R_1 \parallel R_2) \parallel R_4 \quad (21)$$

Part (d) Across the same nodes, 2 and 4, Norton Resistance is same as Thevenin Resistance, $R_{Th} = R_{Nor}$, whereas Norton Current is given by,

$$I_{Nor} = \frac{V_{Th}}{R_{Th/Nor}}$$

$$= \frac{V R_1 (R_2 + R_3 + R_4) + I R_4 R_1 R_2}{R_4 (R_3 R_1 + R_2 R_3 + R_1 R_2)}$$